

Programming Errors

There are two main types of errors in programming:

1. SYNTAX ERRORS

These are errors that are noticed when attempting to compile code – i.e. when you press the green "play button" in Eclipse to run your program. They cause compiling to fail. ***THE PROGRAM WILL NOT RUN!*** The compiler will attempt to identify the error and the line it appears on but you have to go and fix it before you can compile and run your program.

These errors include spelling mistakes, using commands that the compiler does not recognize, forgetting a bracket or a semi-colon, etc. Eclipse is very good at notifying you of syntax errors by placing a "red x" beside any statements that contain a syntax error so you do not even need to wait until compiling your code to discover you have an error!

These are also called "compile-time errors".

2. LOGIC ERRORS

These are errors that cause the compiled program to run differently than intended. They are a result of incorrect programming and NOT incorrect code making them much more difficult to identify and fix. When a program contains logic errors it will run and it will do what you told it to do *but* what you told it to do is wrong! The more complex the problem the more likely logic errors are to appear which is why *it is so important to **plan** your solution and your program BEFORE you begin writing any code.*

These are also called "run-time errors".

Please note: fixing errors in programs is part of the process called **debugging**.